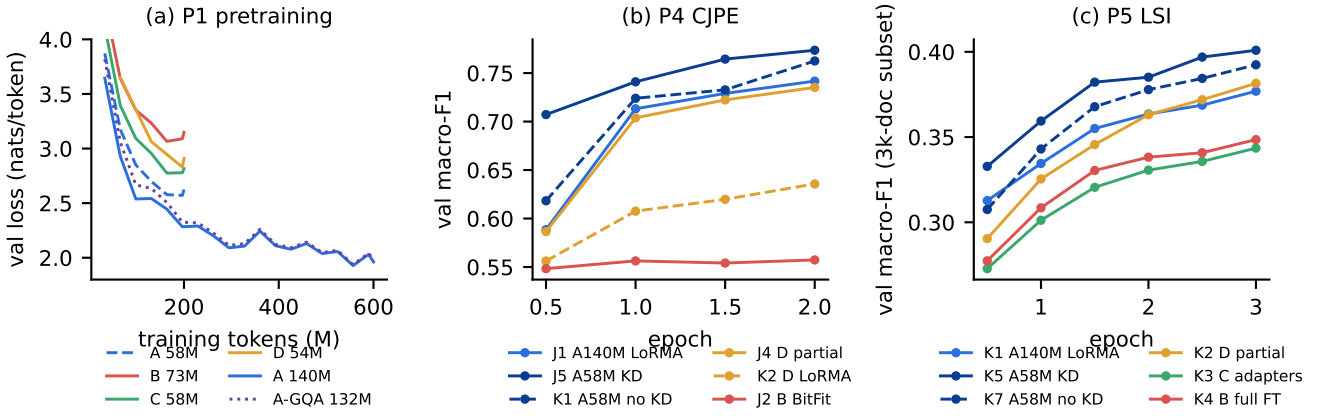
(a) P1 pretraining
val loss (nats/token)
4.0
3.5
3.0
2.5
2.0
200
400
600
training tokens (M)
A 58M
B 73M
C 58M
D 54M
A 140M
A-GQA 132M
(b) P4 CJPE
val macro-F1
0.75
0.70
0.65
0.60
0.55
0.5
1.0
1.5
2.0
epoch
J1 A140M LoRMA
J5 A58M KD
K1 A58M no KD
J4 D partial
K2 D LoRMA
J2 B BitFit
(c) P5 LSI
val macro-F1 (3k-doc subset)
0.40
0.35
0.30
1
2
3
epoch
K1 A140M LoRMA
K5 A58M KD
K7 A58M no KD
K2 D partial
K3 C adapters
K4 B full FT